

Activity:

Investigating the Effect of Saliva on Starch



Aim: To find out how saliva affects starch during digestion.

Tools



Dilute starch solution | Water | Dilute iodine solution | 2 Test tubes | Dropper | Beaker | Warm water (about 40°C) | Cotton swabs | Safety goggles

Steps

1



Use a dropper to put the same amount of dilute starch solution into two test tubes.

2



Soak one cotton swab in saliva and place it in the first test tube. Soak another cotton swab in water and place it in the second test tube.

3



Put both test tubes into a beaker of warm water at about 40°C for a few minutes.

(The warm water provides a temperature similar to that inside our body.)

4



Add 1–2 drops of dilute iodine solution to each test tube.

5



Observe and compare the colors of the two solutions.

Observation



Test tube 1 (Saliva)

- The solution stays yellow-brown.
- This shows that most of the starch has been broken down.



Test tube 2 (Water)

- The solution turns dark blue/black.
- This shows that starch is still present.



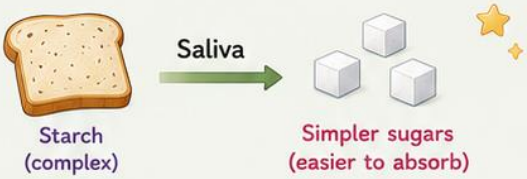
Remember:
Iodine solution gives dark blue/black color when starch is present.

Conclusion



Saliva helps break down starch into simpler sugars during digestion.

The iodine test shows that starch remains in the tube containing water, while much of it has been broken down in the tube containing saliva.



★ **Remember: Saliva helps break down starch into simpler sugars during digestion.**